Predicting Property Tax Value Using Zillow Dataset

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Executive Summary

Project Goal

Predict assessed tax value of properties using Zillow dataset

Approach as a real estate data science problem

Data Preparation

Extensive data cleaning and imputation

Feature engineering to improve data quality

Model Evaluation

Multiple models evaluated including linear regressions and ensemble methods

Gradient Boosting Regressor emerged as top-performing model

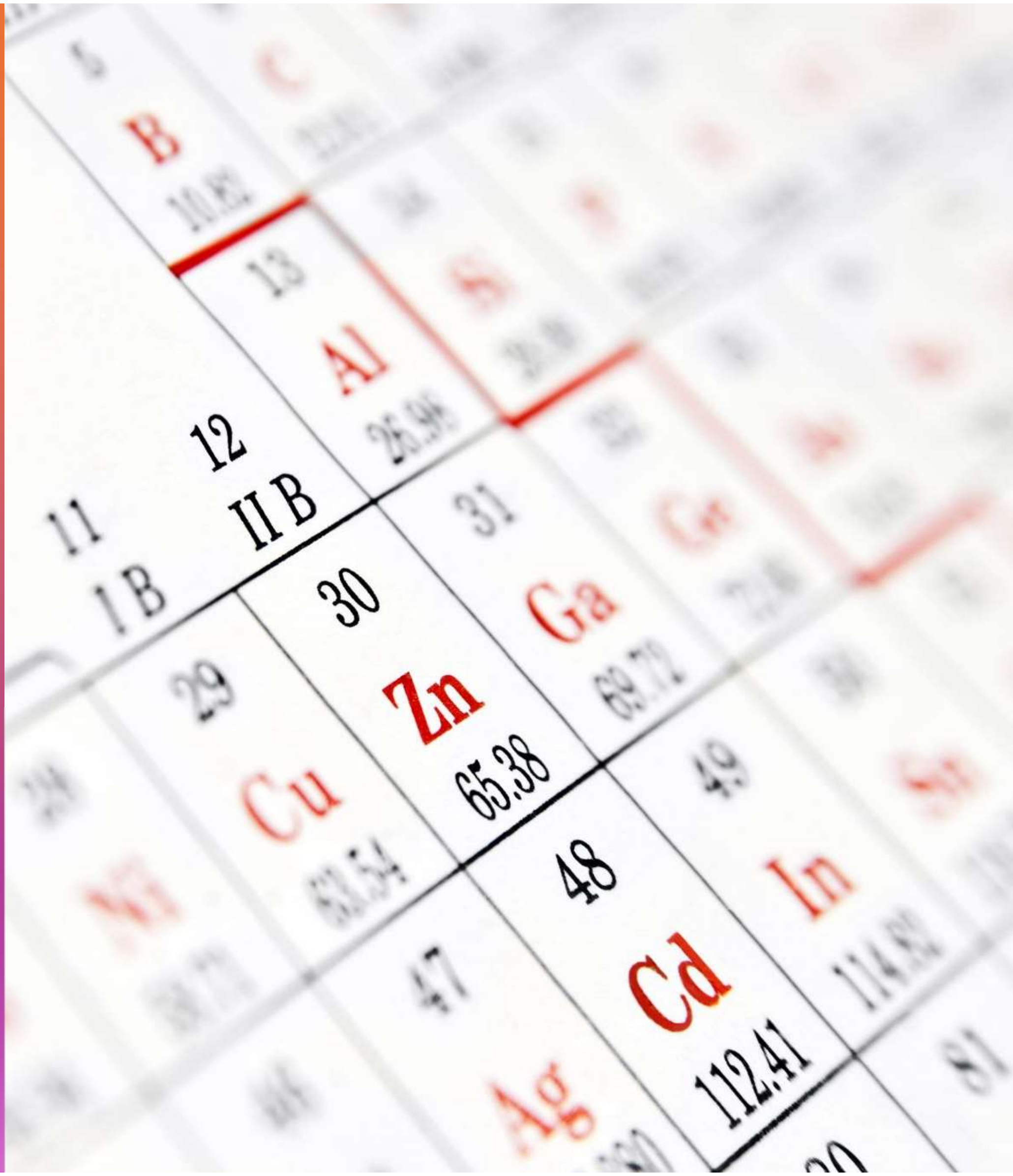
Key Findings

Square footage and location-based variables were most predictive

Business Applications

Introduction

- Importance of Accurate Valuation Models
  - Crucial for internal analytics
  - Essential for customer-facing tools
- Project Objective
  - Enhance accuracy of property tax value predictions
  - Utilize raw Zillow dataset
- Dataset Features
  - Structural features
  - Geographical features
  - Categorical property features
- Data Science Team Tasks
- Primary Business Goals



Data Description

- Dataset Source and Size
  - Sourced from 2017 Zillow Million Dollar Prize dataset by Boston University
  - Contains 77,613 property records
- Features and Variables
  - 55 features including numeric, categorical, and geographic variables
  - Examples: bedroomcnt, yearbuilt, propertycountylandusecode, latitude, longitude
- Target Variable
  - Taxvaluedollarcnt is numeric with a skewed distribution
  - Mean value: $490,151
  - Maximum value: $49M
- Missing Values
- Preprocessing Results

Methodology

- Regression Modeling Framework
  - RMSE used as primary error metric
  - Sensitivity to large errors
- Cross-Validation Technique
  - 5 folds, 5 repeats
  - Ensures reliable performance estimates
  - Mitigates overfitting risks
- Hyperparameter Tuning
  - Guided by validation curves
  - GridSearchCV used
  - Parameters like learning_rate and n_estimators
  - Balances bias and variance

Data Cleaning and Preprocessing

- Issues in Raw Dataset
  - High missingness in features like basementsqft
  - Outliers in taxvaluedollarcnt
- Feature Removal
  - Dropped features with >80% missing values
  - Removed samples with null targets or >50% missing features
- Outlier Removal
  - Applied milder outlier removal (0.5th–99.5th percentiles)
  - Preserved high-value properties
- Imputation Techniques
- Label Encoding
- Adjustments

Feature Engineering

- Non-linear and contextual relationships
  - bath_bed_interaction for livability
  - yearbuilt_squared for non-linear age effects
  - sqft_per_bedroom for space efficiency
  - lat_lon_interaction for location synergy
- EDA insights and domain knowledge
  - High F-values for finishedsquarefeet12
  - Understanding property value drivers
- Feature performance
  - sqft_per_bedroom and yearbuilt_squared retained in top features
  - bath_bed_interaction less impactful, possibly redundant
- Log-transforming the target

Model Selection Process

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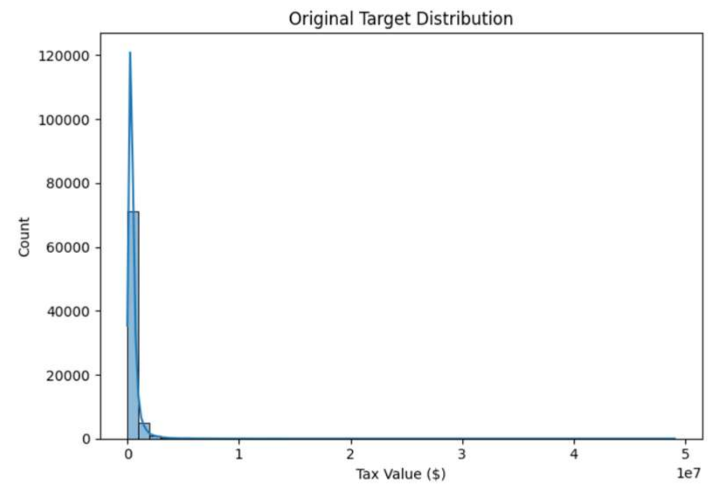
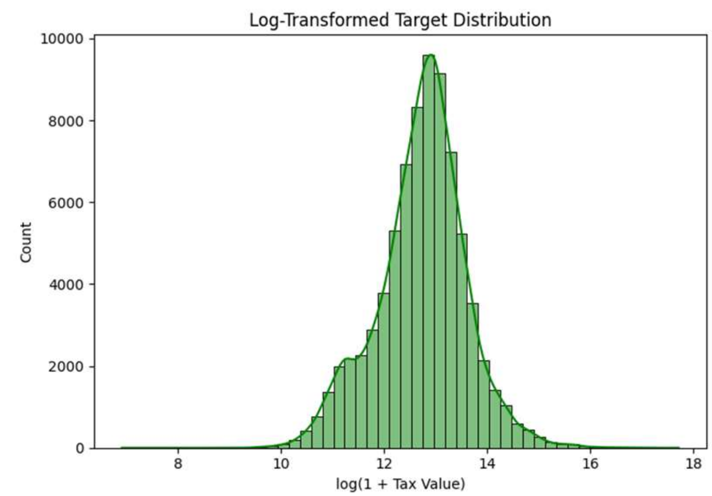
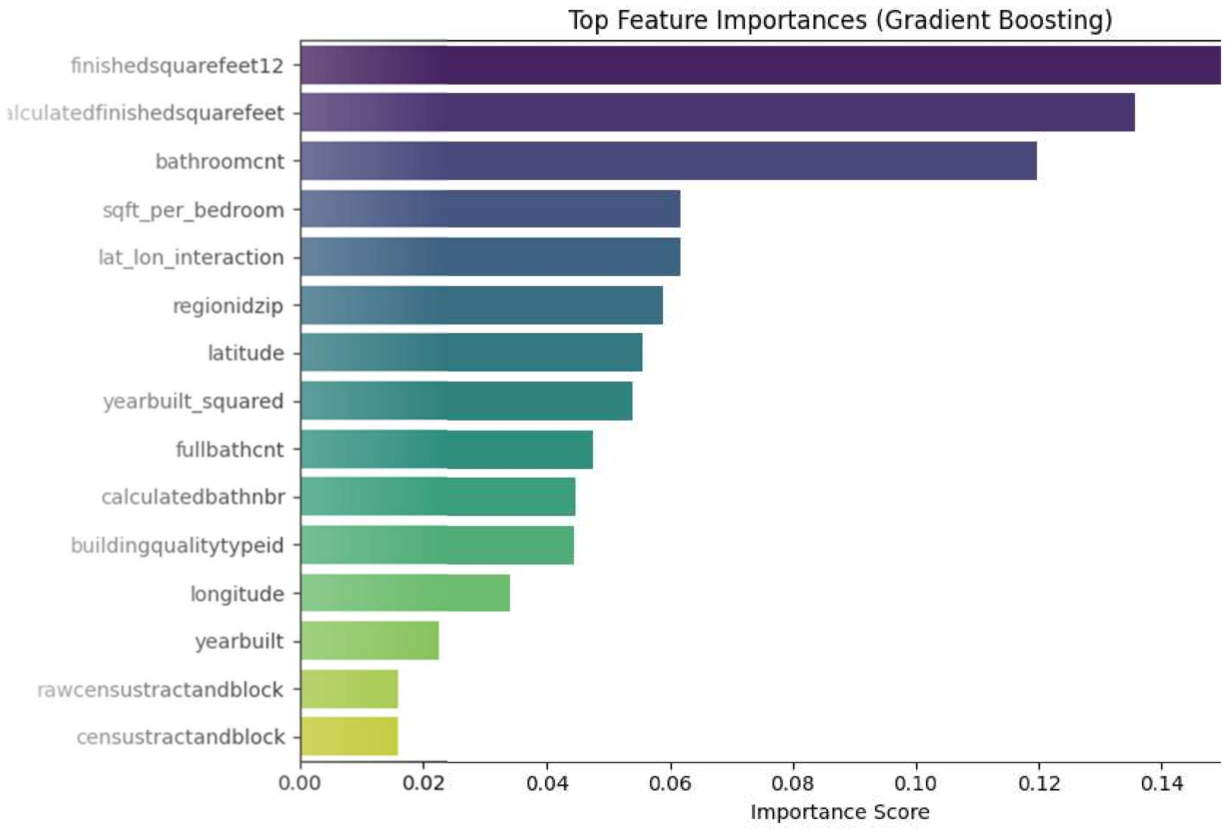
- Model Evaluation
  - Multiple models evaluated including linear regressions and ensemble methods
  - Gradient Boosting Regressor emerged as top-performing model
- Key Findings
  - Square footage and location-based variables were most predictive
- Business Applications





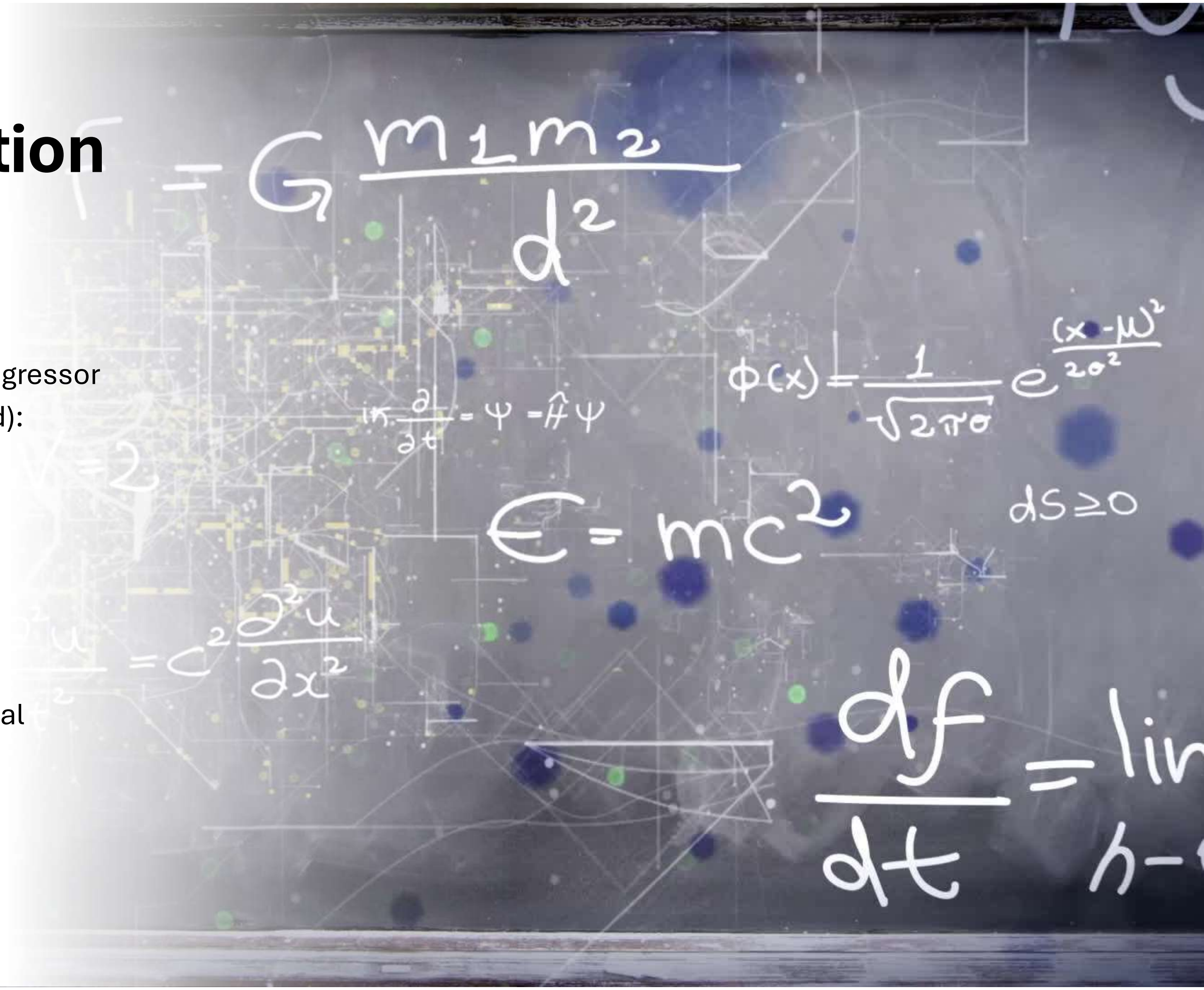
Feature Importance

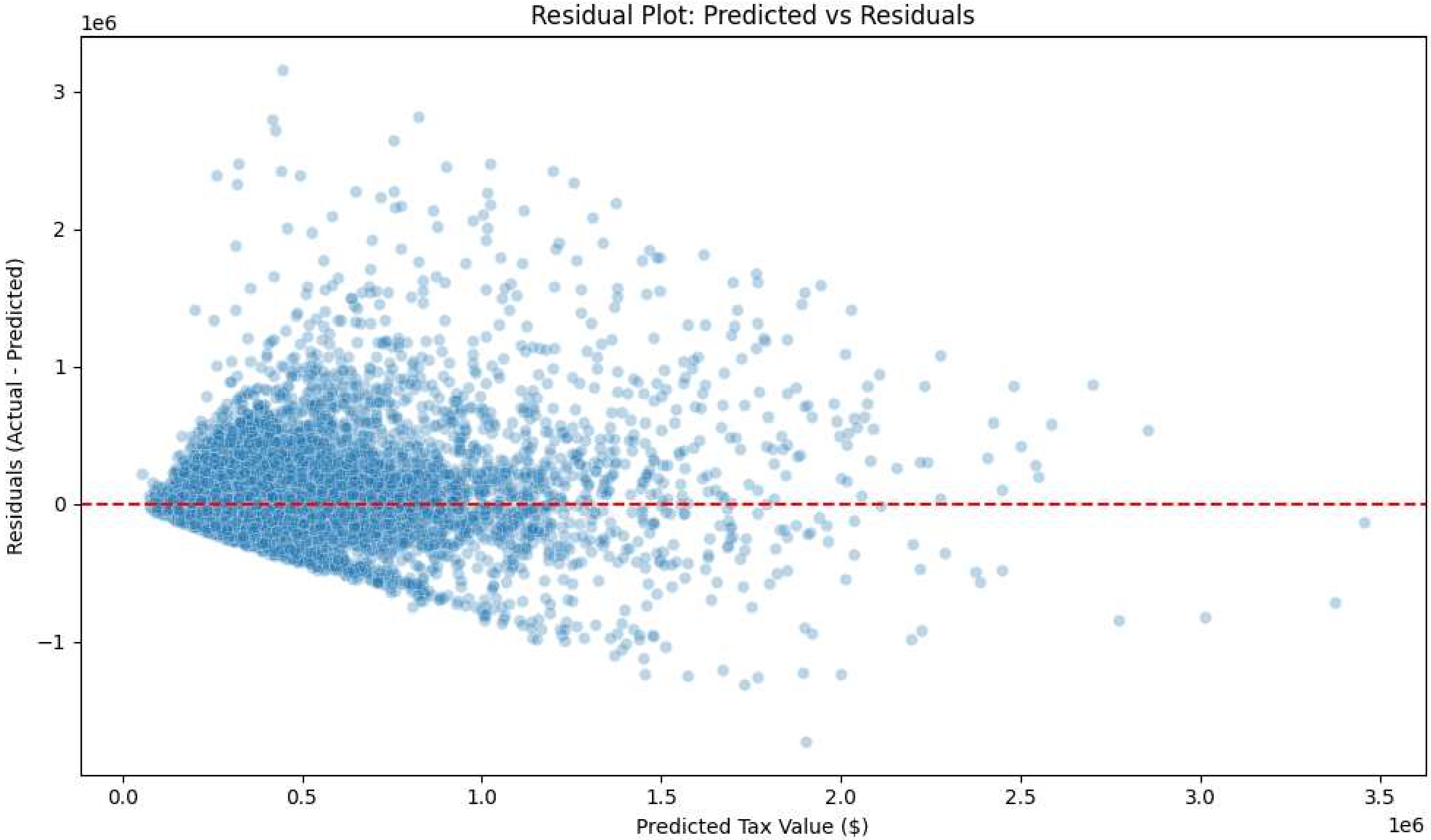
- Key Predictive Features
  - finishedsquarefeet12
  - calculatedfinishedsquarefeet
  - latitude
  - longitude
  - sqft_per_bedroom
- Importance of Square Footage and Location
  - Square footage remains highly predictive
  - Spatial location is a significant factor
- Validation of Engineered Features
  - sqft_per_bedroom ranked high
  - Inclusion of engineered features is validated



Cross-Validation Performance

- Model: Gradient Boosting Regressor
  - RMSE (Log-Transformed): 0.6121 ± 0.0044
  - Parameters:
    - n_estimators: 470
    - learning_rate: 0.1
    - max_depth: 3
    - max_features: 9
- Test Set Performance (Original Scale)
  - Test RMSE: 287,882.76





Error Analysis

- Residuals and Errors
  - Higher errors for properties above $2M
  - Fewer training examples in high-value range
  - Log Transformation helped mitigate skewed distributions and reduce RMSE
- Noise in Zip Codes
  - Less representation in certain zip codes
  - Potential for grouped encoding or smoothing techniques

Conclusion

- Successful Development of Regression Model
  - Achieved test RMSE of $287,882.76
  - Predicts tax-assessed property values
- Workflow Emphasis
  - Strong preprocessing
  - Engineered feature design
  - Thoughtful model selection
- Foundation for Zillow-like Systems
  - Supports better pricing strategies
  - Enhances consumer tools
- Limitations
- Future Improvements